

**INSTITUTE OF ACTUARIES OF INDIA**

**EXAMINATIONS**

**May 2026**

**SA7 - Investment and Finance**

**Time allowed: 3 Hours 15 Minutes**

**Total Marks: 100**

**Q.1)** An Indian life insurance company, "Apex Life," is navigating a volatile macroeconomic environment characterized by protracted global geopolitical conflicts. This has resulted in sustained supply-chain disruptions and structurally higher global energy and food prices. Consequently, the Reserve Bank of India (RBI) has maintained a prolonged tight monetary stance. Furthermore, recent macroeconomic data highlights a widening Current Account Deficit (CAD), exacerbated by surging seasonal gold imports.

Simultaneously, under revised IRDAI investment and finance guidelines, Apex Life is overhauling its Asset-Liability Management (ALM) and redesigning its distributor compensation. To retain top-tier agents, the company proposes a deferred "Loyalty Commission" payable at the end of policy year 5. This payout will be explicitly indexed: 50% to the Indian Consumer Price Index (CPI) and 50% to the Nifty 50 Total Return Index.

Additionally, Apex Life is setting up an International Financial Services Centre Insurance Office (IIO) in GIFT City under the IFSC regulatory framework to optimize its capital and investment strategies.

**a)** The geopolitical conflict and shifting macroeconomic variables necessitate a re-evaluation of traditional asset class behaviours.

**i)** Define the expected changes in the correlation between domestic sovereign bonds, equities, and gold in response to an energy-driven stagflationary shock. In your analysis, incorporate the transmission mechanisms of a widening trade deficit and CAD pressures on these asset classes. **[6]**

**ii)** Discuss the specific impact of this correlation breakdown on the solvency position of a life insurer writing primarily guaranteed non-participating savings products. In your answer, explicitly consider how the specific features of these products—such as fixed long-term nominal cash outflows, the inability to pass on capital losses to policyholders, and guaranteed surrender value floors—affect the net balance sheet when traditional asset diversification fails. **[6]**

**b)** The Apex Life Investment Committee must revise its SAA in light of the geopolitical conflict.

Explain how the behavioral biases of Availability Heuristic and Loss Aversion might lead the committee to make sub-optimal asset allocation decisions in the current environment. Propose three specific governance mechanisms the Chief Investment Officer (CIO) should implement to counteract groupthink and enforce rational decision-making, ensuring compliance with the independence and control function mandates of IRDAI's corporate governance guidelines. **[10]**

**c)** The new deferred "Loyalty Commission" creates a complex liability for Apex Life.

**i)** Characterize the nature of this liability from an ALM perspective, explicitly identifying the embedded optionality if the commission structure includes a "zero-floor" (i.e., the commission cannot be negative). **[4]**

**ii)** Design a practical replicating portfolio using domestic Indian financial instruments to hedge this liability, considering how these hedging instruments would be classified and measured under Ind AS 109. **[5]**

iii) Critically evaluate the practical limitations and basis risks associated with your proposed hedge, particularly concerning the CPI-linked component. [4]

d) The Board intends to utilize the new GIFT City IIO for capital management and global investment expansion. The International Financial Services Centre (IFSC) in GIFT City operates as a distinct offshore jurisdiction within India, offering a separate regulatory regime designed to facilitate global capital flows, free from many domestic investment constraints and currency controls.

i) Based on the premise that an IFSC operates as an offshore zone, discuss the strategic advantages regarding permissible asset classes (such as global alternatives and foreign derivatives) and currency diversification compared to the heavily regulated Domestic Tariff Area (DTA). [4]

ii) Propose a Financial Reinsurance (FinRe) structure executed between the DTA entity and the IIO designed to monetize the domestic book's Value-in-Force (VIF)—defined here as the present value of unrecognized, future expected profits embedded in existing long-term policies. Explain the mechanics of how this transaction converts future profits into immediate cash, thereby improving the domestic Available Solvency Margin. [4]

iii) Assess the regulatory risks and potential conflicts of interest inherent in related-party FinRe transactions across differing regulatory regimes. [3]

iv) Drawing on recent central bank developments, discuss how the IIO could leverage tokenised assets on a Unified Ledger and the Wholesale Digital Rupee (e-Rupee) to optimize cross-border settlement and liquidity management. [4]

**[50 marks]**

**Q.2)** "Pinnacle Financial Services," a large Indian financial conglomerate, is in the final stages of acquiring "Heritage Life," a mid-sized, traditional life insurer. Heritage Life's portfolio is predominantly invested in domestic government securities and highly-rated corporate bonds, backing a large block of non-participating guaranteed products and a closed block of legacy Unit-Linked Insurance Plans (ULIPs) with "Highest Net Asset Value (NAV)" guarantees.

Pinnacle's strategic objective post-acquisition is to modernize Heritage Life's investment function. This includes a major allocation to Alternative Investment Funds (AIFs) to harvest illiquidity premiums. Furthermore, the combined entity must navigate an upcoming accounting transition to a market-consistent liability valuation framework and a shift to a Risk-Based Capital (RBC) regime.

a) As part of the Mergers and Acquisitions (M&A) due diligence, Pinnacle is valuing Heritage Life's asset portfolio. Heritage Life holds a stressed portfolio of real estate corporate bonds and bank Perpetual Debt Instruments (Additional Tier 1 (AT1) bonds).

i) Explain the challenges in establishing "Fair Value" for these distressed assets. In your answer, contrast "Mark-to-Market" with a "Mark-to-Model" Expected Credit Loss (ECL) approach. Given a recent tightening in domestic bankruptcy laws that requires a 90% creditor approval to withdraw insolvency applications (hindering out-of-court restructuring), and new banking regulations that mandate permanent principal write-downs or coupon cancellations for AT1 bonds at the point of non-viability, discuss how these specific regulatory shifts impact the recovery assumptions used in your model. [8]

- ii) Critically evaluate how the transition to a new market-consistent accounting standard—one that replaces historic cost with current fair value for both assets and liabilities—alongside a new Risk-Based Capital (RBC) framework—which replaces flat capital requirements with explicit, granular capital charges for credit and ALM mismatch risks—alters the "Cost of Capital" and investment volatility assumptions used in valuing the target company during an acquisition. [4]
- b) Pinnacle's CIO plans to allocate ₹1,000 crores of the free surplus into Indian Private Equity (PE) and Venture Capital (VC) funds.
- i) Traditional manager selection relies heavily on past performance. Discuss why relying on historical Sharpe Ratios and past track records is insufficient and potentially misleading when selecting a PE fund manager. [3]
- ii) Propose a comprehensive due diligence framework for selecting a PE manager. Ensure you explicitly contrast the permissibility and characteristics of Category I, Category II, and Category III AIFs under the SEBI (AIF) Regulations, 2012 from the perspective of a life insurance investor. [6]
- iii) Explain the "J-Curve" effect in private equity. Identify two performance metrics appropriate for evaluating a PE fund during its lifecycle, and explain their limitations. [4]
- c) The legacy block of ULIPs poses a significant risk to the newly merged entity's balance sheet under an RBC regime due to its "Highest NAV" guarantees. (For the purposes of risk mitigation and control, this "Highest NAV" guarantee should be treated strictly as an embedded exotic financial derivative independent of any specific life insurance product expertise)
- i) Define "Delta" and "Gamma" risks in the context of investment guarantees embedded within life insurance products. [3]
- ii) Outline a "Dynamic Hedging" strategy using equity index futures and options to manage this guarantee risk, ensuring the strategy complies with IRDAI's derivative policy guidelines. [6]
- iii) Discuss the practical operational and market risks associated with running a dynamic hedging program, specifically highlighting the impact of "transaction costs" and "gap risk" during market crashes. [4]
- d) Following the acquisition, Pinnacle must integrate Heritage Life into its group-wide Performance Measurement and Attribution framework.
- i) Describe the Brinson-Fachler attribution model. Explain how it separates the value added by the portfolio manager into "Asset Allocation" and "Security Selection" effects. [6]
- ii) Discuss the specific challenges the performance measurement team will face when attempting to benchmark and attribute the performance of the newly added Private Credit and Private Equity portfolios against traditional, liquid market indices. [6]

[50 marks]

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